

Introduction to a Web of Linked Data

Week 1: Principles of a Web of Linked Data

WEEK 1: Principles of a Web of Linked Data

1. Web architecture
2. Separating presentation and content
3. From pages to resources
4. Linked data principles
5. Stack of standards and languages

WEEK 1: Principles of a Web of Linked Data

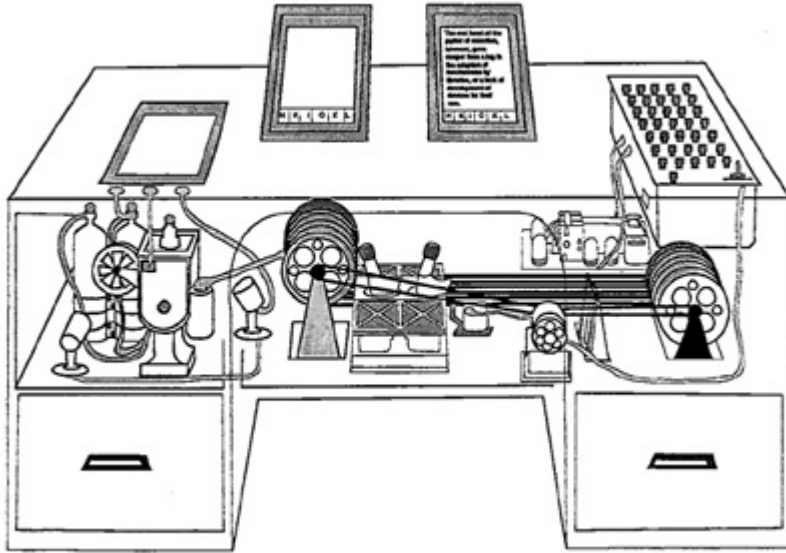
1. Web architecture
2. Separating presentation and content
3. From pages to resources
4. Linked data principles
5. Stack of standards and languages

Externalized Linking



- Vannevar BUSH, 1890-1974

Externalized Linking



Memex, Life Magazine, 10 September 1945



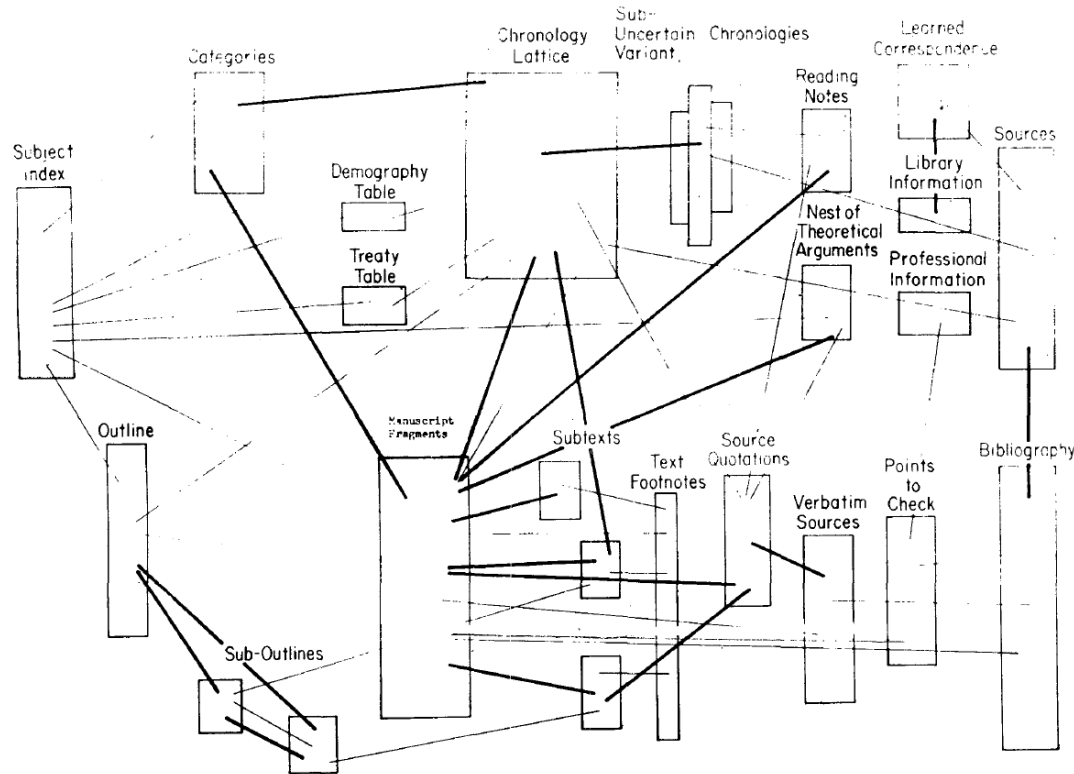
- Vannevar BUSH, 1890-1974
- Memex = Memory Extension
 - External management of links
 - Assist recall

Linking Digital Documents



- Ted Nelson

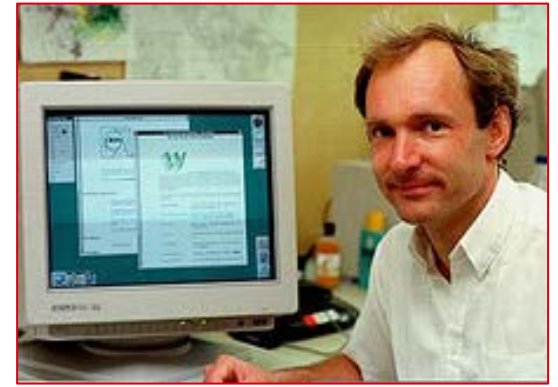
Linking Digital Documents



- Ted Nelson
- Hypertext
- Hypermedia

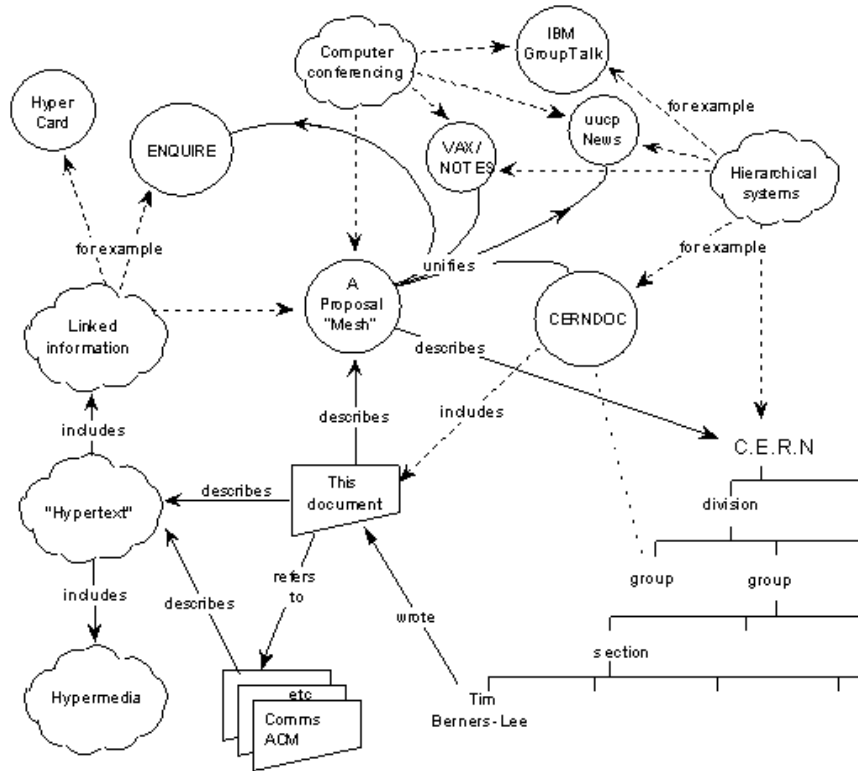
Complex information processing: a file structure for the complex, the changing and the indeterminate, T. H. Nelson, ACM, 1965

Linking Across Networks



- Tim Berners-Lee

Linking Across Networks

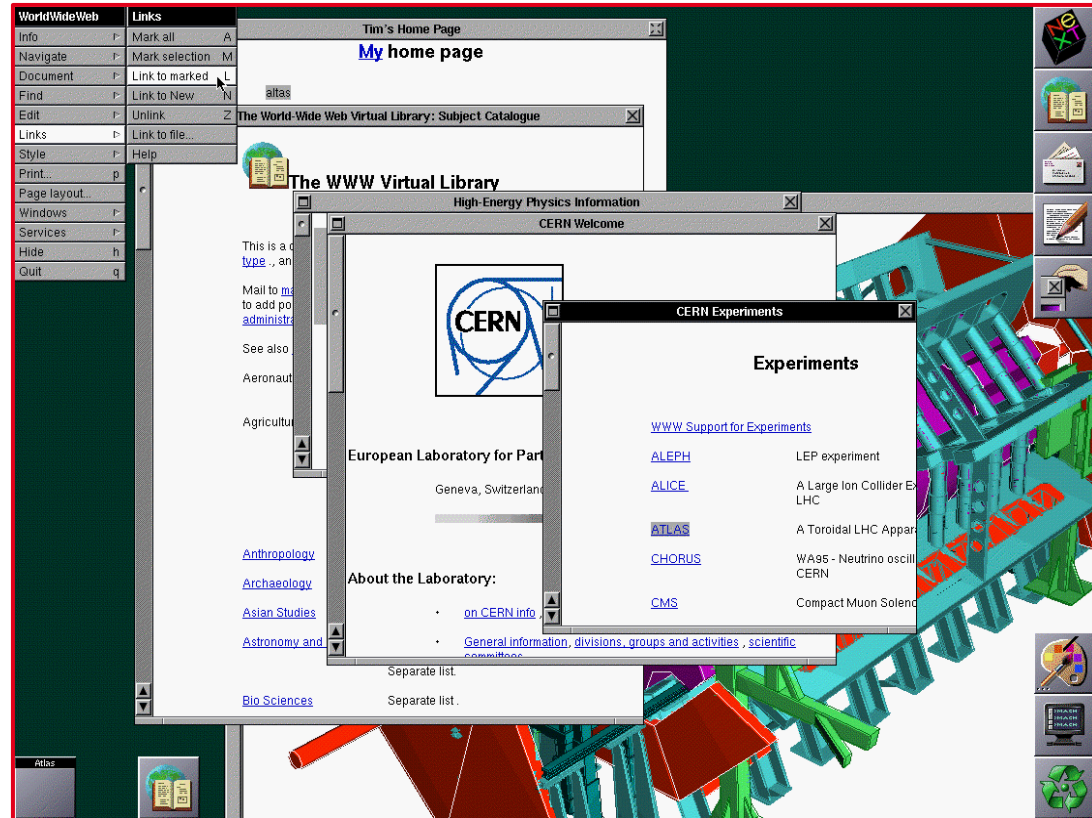


- Tim Berners-Lee
- Identify and link across Internet

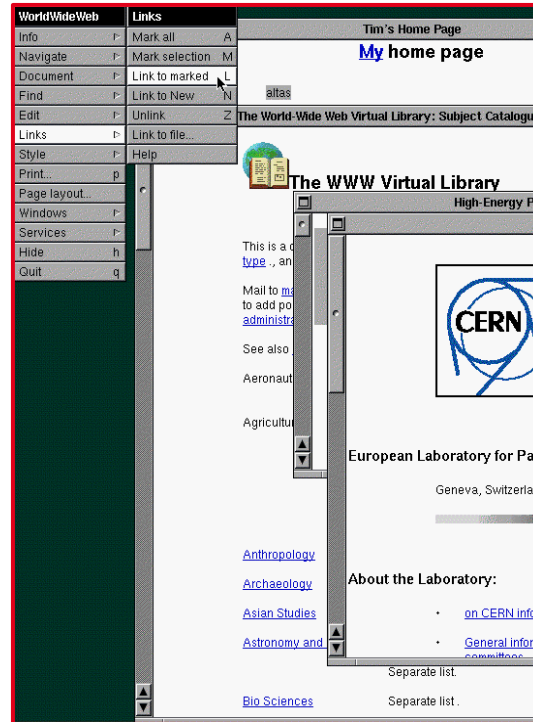
Information Management: A Proposal

Tim Berners-Lee, CERN, March 1989, May 1990

Architecture of the Web



Architecture of the Web



client
(browser)



server

Three Components of the Web Architecture

1. Identification (URI) & addressing (URL)
ex. `http://www.inria.fr`



Three Components of the Web Architecture

1. Identification (URI) & addressing (URL)

ex. `http://www.inria.fr`



2. Communication / protocol (HTTP)

`GET /centre/sophia HTTP/1.1`

`Host: www.inria.fr`



Three Components of the Web Architecture

1. Identification (URI) & addressing (URL)

ex. `http://www.inria.fr`



2. Communication / protocol (HTTP)

`GET /centre/sophia HTTP/1.1`

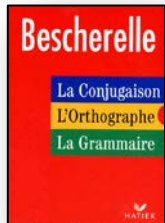
`Host: www.inria.fr`



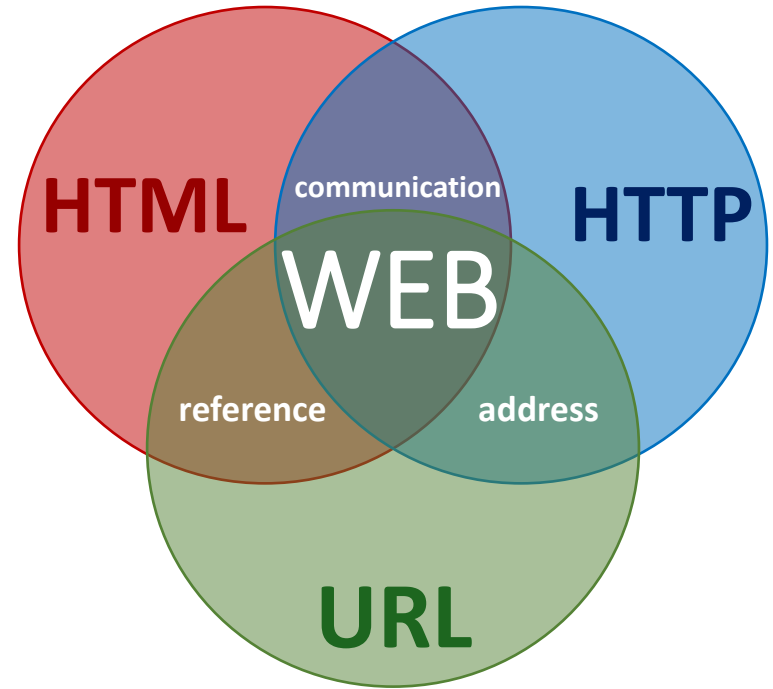
3. Language of representation (HTML)

Fabien works at

`<a href="http://inria.fr">Inria</a>`



Three Components of the Web Architecture



Picture credits

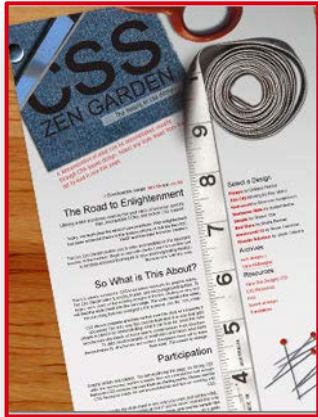
- Vannevar Bush with his Differential Analyzer, c. 1935. MIT Museum by Revol Web - CC BY-SA 2.0 <https://www.flickr.com/photos/revolweb/3985110824/>
- Memex, Life Magazine, 10 September 1945
- Ted Nelson by Gisle Hannemyr - CC-BY-SA-3.0
- Complex information processing: a file structure for the complex, the changing and the indeterminate, T. H. Nelson, ACM, 1965
- Tim Berners-Lee by Athanasios Kasampalis - CC BY-NC-ND 2.0 <https://www.flickr.com/photos/12693492@N04/1339026964>
- Information Management: A Proposal Tim Berners-Lee, CERN, March 1989, May 1990
- Screenshot of the original web browser NeXT in 1993 CERN - <http://home.web.cern.ch/fr/about/updates/2013/04/twenty-years-free-open-web>
- Server icon, Octicons by Github - MIT License www.iconfinder.com
- EAN by Crevette - CC BY-NC-SA 2.0 <https://flic.kr/p/4snzAi>
- Tin can phone by Kit Cowan - CC BY-NC-ND 2.0 <https://flic.kr/p/4cUMvV>
- Bescherelle © Editions Hatier <http://desencyclopediae.wikia.com/wiki/Fichier:Bescherelle.gif>

WEEK 1: Principles of a Web of Linked Data

1. Web architecture
- 2. Separating presentation and content**
3. From pages to resources
4. Linked data principles
5. Stack of standards and languages

Separating Presentation and Content (CSS)

CSS Zen Garden: one HTML content but a collection of cascading style sheets



Propose your Own Languages (XML)



structure data



using tags



in a textual format

Propose your Own Languages (XML)



structure data



using tags



in a textual format



open standard family of languages



combine different languages



open and non-proprietary

XML 101

`<root>...</root>`

`<x>...</x>` or `<x/>`

`<a><b></a></b>` 

`<x>` $\neq$ `<X>`

`<1an>` `<xmla>` `<bla bla>` 

`<a b='c'></a>` or `<a b='c' />`

Example of a business card

```
<card>  
  <name>gandon</name>  
  <tel type="office">+33492965170</tel>  
  <page url="fabien.info"/>  
</card>
```

Root

<card>

<name>gandon</name>

<tel type="office">+33492965170</tel>

<page url="fabien.info"/>

</card>

Opening and closing tags

```
<card>  
  <name>gandon</name>  
  <tel type="office">+33492965170</tel>  
  <page url="fabien.info"/>  
</card>
```

Attributes

```
<card>  
  <name>gandon</name>  
  <tel type="office">+33492965170</tel>  
  <page url="fabien.info"/>  
</card>
```

Self-closing tag

```
<card>  
  <name>gandon</name>  
  <tel type="office">+33492965170</tel>  
  <page url="fabien.info"/>  
</card>
```



Namespace: Grouping the Terms Used to Structure

Definition: abstract space gathering names in a same set

<http://inria.fr/topics#compilation>

<http://mit.edu/org/Lab>

e.g. a dictionary, a library index, a glossary, a standard, a thesaurus,...

NAMESPACES,

- A collection of names identified by a URI
- Names belonging to a same namespace start with the same URI

PREFIXES,

- Local shortcuts to declare a namespace in a file, document, etc.
- Use prefixes locally instead of repeating the namespace
- Representation languages provide prefix declaration means

AND QUALIFIED NAMES

- prefix + ":" + local name
- e.g. dc:title instead of <http://purl.org/dc/elements/1.1/title>

Example of a namespace declaration

```
<e:card xmlns:e="http://ns.inria.fr/ex/">  
  <e:name>gandon</e:name>  
  <e:tel e:type="office">+33492965170</e:tel>  
  <e:page e:url="fabien.info"/>  
</e:card>
```

Example of namespace usage

```
<e:card xmlns:e="http://ns.inria.fr/ex/">  
  <e:name>gandon</e:name>  
  <e:tel e:type="office">+33492965170</e:tel>  
  <e:page e:url="fabien.info"/>  
</e:card>
```

XML Gave Birth to a Diversity of Languages

MusicML MathML SVG (graphics)
OpenOffice (documents)
XACML (access) FpML (finance)
SMIL (multimedia) CML
(chemistry) GPX (GPS) WSDL
(services) NewsML XHTML ...

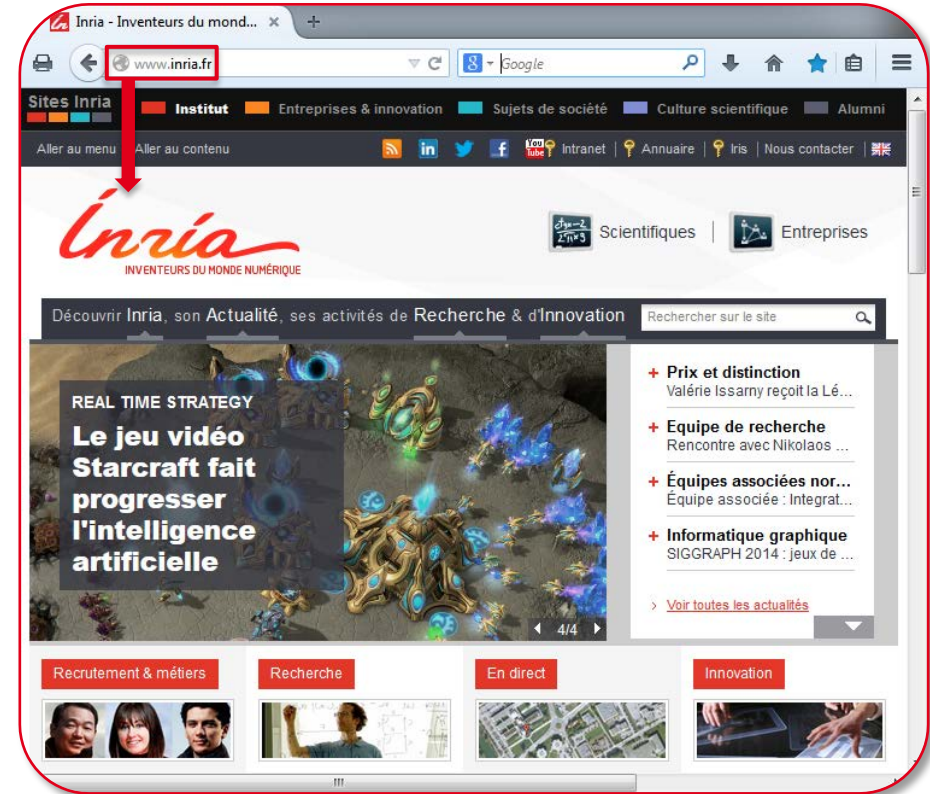
Picture credits

- CSS Zen Garden by Dave Shea - CC BY-NC-SA 3.0
<http://www.csszengarden.com/>
 - ✓ Lyli Pond, Rose Thorogood, Australia <http://www.csszengarden.com/201>
 - ✓ Peace of Mind, Carlos Varela, Chile <http://www.csszengarden.com/184>
 - ✓ Wiggles the Wonderworm, Joseph Pearson, Australia
<http://www.csszengarden.com/099>
 - ✓ A Simple Sunrise, Rob Soule, United States <http://www.csszengarden.com/158>
 - ✓ Jackie Caradonio <http://www.jackiecaradonio.com/ZenGarden/FINALb.html>
 - ✓ Metacopsunlimited
<http://metacopsunlimited.com/mc/index.php/MCv4/combinedFolio>
- XML in 10 points, W3C <http://www.w3.org/XML/1999/XML-in-10-points.html.en>

WEEK 1: Principles of a Web of Linked Data

1. Web architecture
2. Separating presentation and content
- 3. From pages to resources**
4. Linked data principles
5. Stack of standards and languages

Locate a Page (URL)



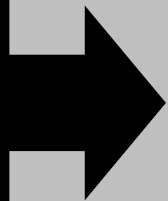
URL

identify what exists on
the Web

`http://my-site.fr`



URL



URI

identify what exists on
the Web

identify,
on the Web,
what exists

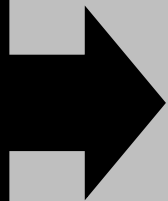
<http://my-site.fr>



<http://animals.org/zebra#this>



URL



URI



IRI

identify what exists on
the Web

identify,
on the Web,
what exists

identify,
on the Web,
in any language,
what exists

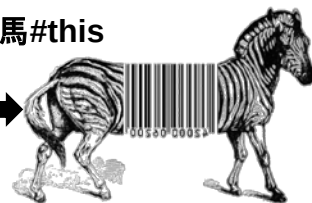
<http://my-site.fr>



<http://animals.org/zebra#this>



<http://الحيوانات.tn/斑馬#this>



URL → URI → IRI

RESOURCE

The diagram illustrates the relationship between three types of identifiers: URL, URI, and IRI. The letters 'U', 'R', and 'I' are shown in a light gray font, while 'L', 'U', 'R', and 'I' are in a bold black font. A dashed arrow points from 'L' to 'U', and another dashed arrow points from 'R' to 'I'. Below the text, two horizontal arrows indicate that 'RESOURCE' is a subset of both 'URI' and 'IRI'.

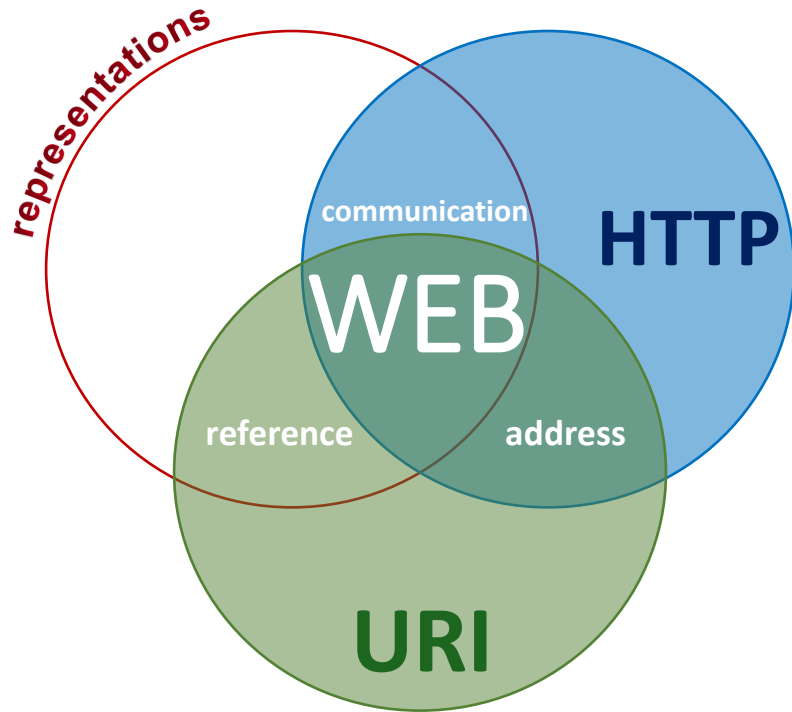


Definition: a resource is anything that can be identified by a URI

<http://fabien.info/objects#mycar>

e.g. a page, a person, a car, a dog, an idea, a country, a product, a service...

Beyond Documentary Representations



Picture credits

- Inria website <http://www.inria.fr>
- Barcode Artwork, Barcodes Limited <https://barcode1.ie/wp-content/assets/sites/5/Stretch-Zebra-barcode.jpg>
- Icons (iconfinder.com):
 - Faq, help, mark, query, question, support icon by ibrandify – Free for commercial use
 - Global, globe, internet icon, iOS 7 icons by Visual Pharm – CC BY ND 3.0

WEEK 1: Principles of a Web of Linked Data

1. Web architecture
2. Separating presentation and content
3. From pages to resources
- 4. Linked data principles**
5. Stack of standards and languages

A Web Approach to Data Publication

1. Use HTTP URI (URL) to allow dereferencing the address
e.g. `http://researcher.fr/fgandon`

A Web Approach to Data Publication

1. Use HTTP URI (URL) to allow dereferencing the address
e.g. `http://researcher.fr/fgandon`
2. When a URI is accessed, provide data about the resource it represents (HTTP)

A Web Approach to Data Publication

1. Use HTTP URI (URL) to allow dereferencing the address
e.g. <http://researcher.fr/fgandon>
2. When a URI is accessed, provide data about the resource it represents (HTTP)
3. Include in these data links to other data (Web)

A Web Approach to Data Publication



1. Use HTTP URI (URL) to allow dereferencing the address
e.g. <http://researcher.fr/fgandon>
2. When a URI is accessed, provide data about the resource it represents (HTTP)
3. Include in these data links to other data (Web)

A Web Approach to Data Publication



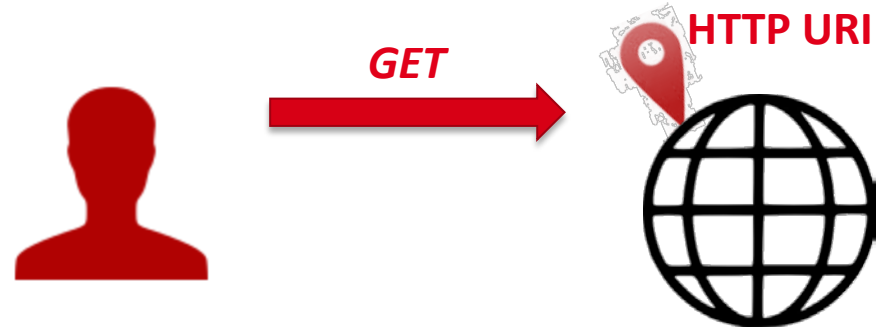
URI ???...



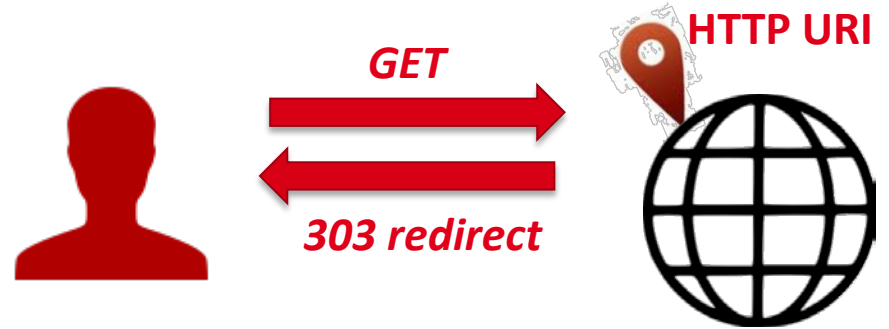
A Web Approach to Data Publication



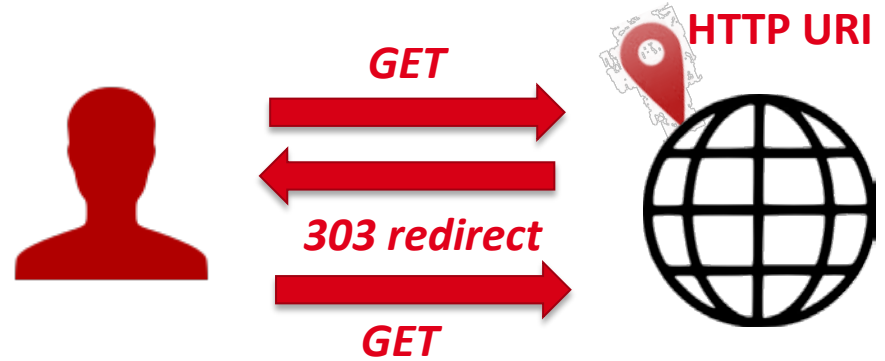
A Web Approach to Data Publication



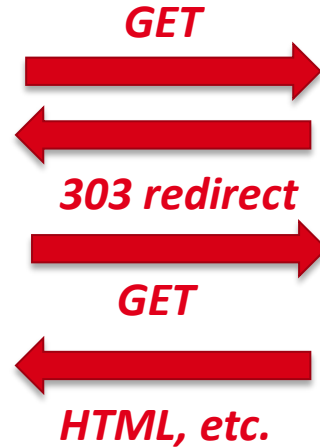
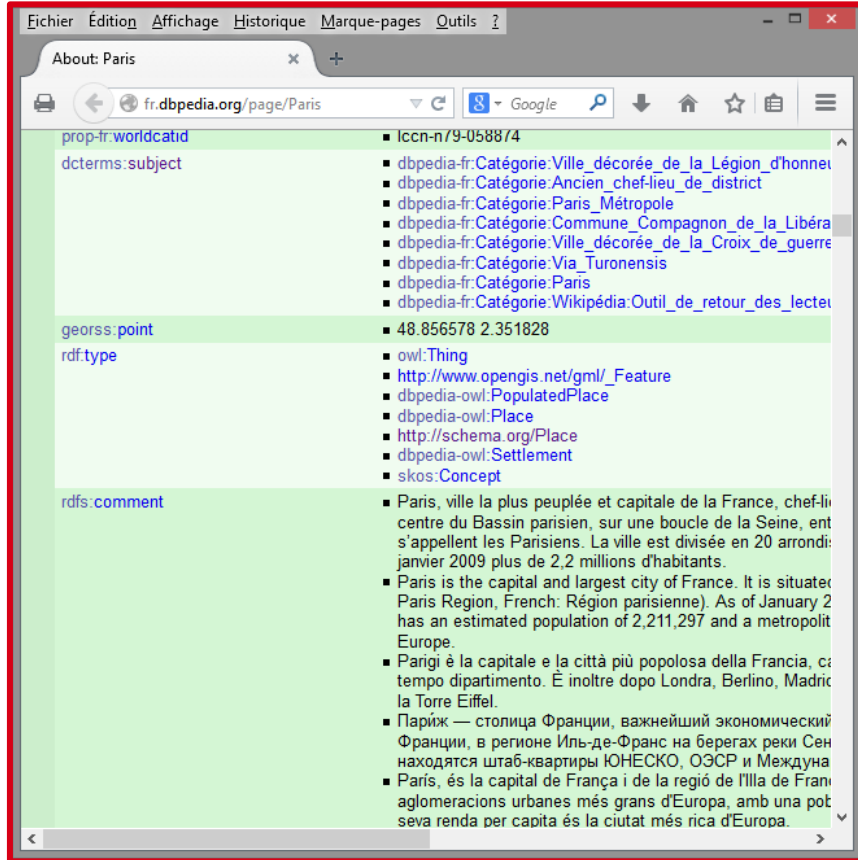
A Web Approach to Data Publication



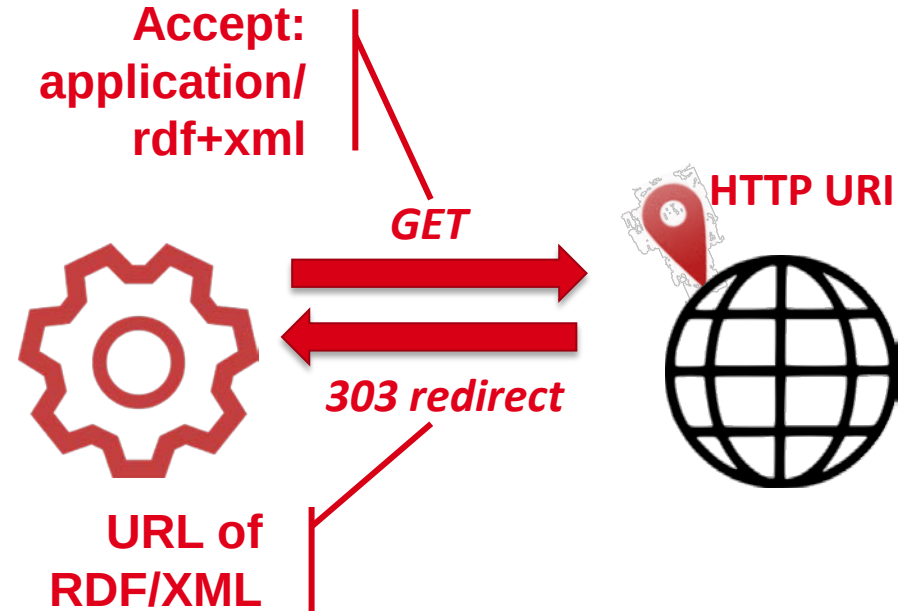
A Web Approach to Data Publication



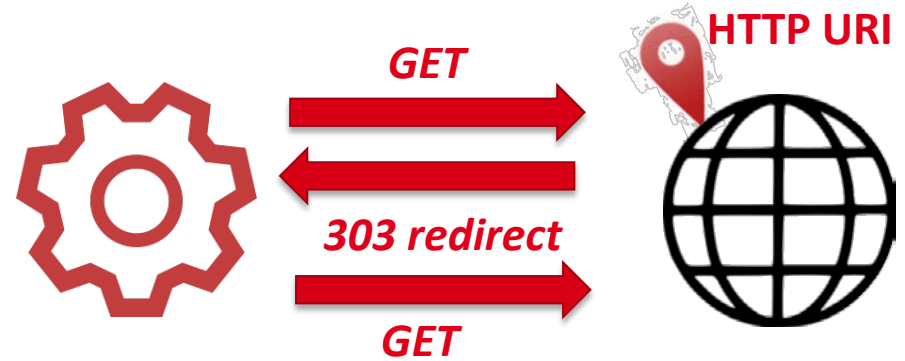
a Web approach to data publication



A Web Approach to Data Publication (conneg)



A Web Approach to Data Publication (conneg)



A Web Approach to Data Publication (conneg)

```
Fichier Édition Affichage Historique Marque-pages Outils ?
http://fr.dbpedia.org/data/Paris.rdf

<rdf:Description rdf:about="http://fr.dbpedia.org/resource/
/Liste_de_périphrases_désignant_des_pays">
  <dbpedia-owl:wikiPageWikiLink rdf:resource="http://fr.dbpedia.org/resource/Paris"/>
</rdf:Description>
<rdf:Description rdf:about="http://fr.dbpedia.org/resource/Stéphane_Mallarmé">
  <dbpedia-owl:birthPlace rdf:resource="http://fr.dbpedia.org/resource/Paris"/>
  <dbpedia-owl:wikiPageWikiLink rdf:resource="http://fr.dbpedia.org/resource/Paris"/>
</rdf:Description>
<rdf:Description rdf:about="http://fr.dbpedia.org/resource/
/Système_international_d'unités">
  <dbpedia-owl:wikiPageWikiLink rdf:resource="http://fr.dbpedia.org/resource/Paris"/>
</rdf:Description>
<rdf:Description rdf:about="http://fr.dbpedia.org/resource/Sharon_Stone">
  <dbpedia-owl:wikiPageWikiLink rdf:resource="http://fr.dbpedia.org/resource/Paris"/>
</rdf:Description>
<rdf:Description rdf:about="http://fr.dbpedia.org/resource/Seine">
  <prop-fr:villes rdf:resource="http://fr.dbpedia.org/resource/Paris"/>
  <dbpedia-owl:city rdf:resource="http://fr.dbpedia.org/resource/Paris"/>
  <dbpedia-owl:wikiPageWikiLink rdf:resource="http://fr.dbpedia.org/resource/Paris"/>
</rdf:Description>
<rdf:Description rdf:about="http://fr.dbpedia.org/resource/Seconde_Guerre_mondiale">
  <dbpedia-owl:wikiPageWikiLink rdf:resource="http://fr.dbpedia.org/resource/Paris"/>
</rdf:Description>
<rdf:Description rdf:about="http://fr.dbpedia.org/resource/Stanley_Kubrick">
  <dbpedia-owl:wikiPageWikiLink rdf:resource="http://fr.dbpedia.org/resource/Paris"/>
</rdf:Description>
<rdf:Description rdf:about="http://fr.dbpedia.org/resource/Sigmund_Freud">
```



A Web Approach to Data Publication (conneg)

Fichier Édition Affichage Historique Marque-pages Outils ?

About: Paris

fr.dbpedia.org/page/Paris

prop-fr:worldcatid
dcterms:subject

- lccn-n/9-058874
- dbpedia-fr:Catégorie:Ville_décorée_de_la_Légion_d'honneur
- dbpedia-fr:Catégorie:Ancien_chef-lieu_de_district
- dbpedia-fr:Catégorie:Paris_Métropole
- dbpedia-fr:Catégorie:Commune_Compagnon_de_la_Libération
- dbpedia-fr:Catégorie:Ville_décorée_de_la_Croix_de_guerre
- dbpedia-fr:Catégorie:Via_Turonensis
- dbpedia-fr:Catégorie:Paris
- dbpedia-fr:Catégorie:Wikipédia:Outil_de_retour_des_lecteurs

georss:point
rdf:type

- 48.856578 2.351828
- owl:Thing
- http://www.opengis.net/gml/_Feature
- dbpedia-owl:PopulatedPlace
- dbpedia-owl:Place
- http://schema.org/Place
- dbpedia-owl:Settlement
- skos:Concept

rdfs:comment

- Paris, ville la plus peuplée et capitale de la France, chef-lieu du centre du Bassin parisien, sur une boucle de la Seine, est surnommée la Ville Lumière. La ville est divisée en 20 arrondissements. En janvier 2009 plus de 2,2 millions d'habitants.
- Paris is the capital and largest city of France. It is situated in the Paris Region, French: Région parisienne). As of January 2009, it has an estimated population of 2,211,297 and a metropolitan area of 10,781,000.
- Parigi è la capitale e la città più popolosa della Francia, e fa parte del dipartimento. È inoltre dopo Londra, Berlino, Madrid e Roma la città più visitata del mondo.
- Пари́ж — столица Франции, важнейший экономический центр страны. В регионе Иль-де-Франс на берегах реки Сена находятся штаб-квартиры ЮНЕСКО, ОЭСР и Международного валютного фонда.
- Paris, és la capital de França i de la regió de l'Illa de França. És una de les aglomeracions urbanes més grans d'Europa, amb una població de 2.211.297 habitants. La seva renda per capita és la ciutat més rica d'Europa.

Fichier Édition Affichage Historique Marque-pages Outils ?

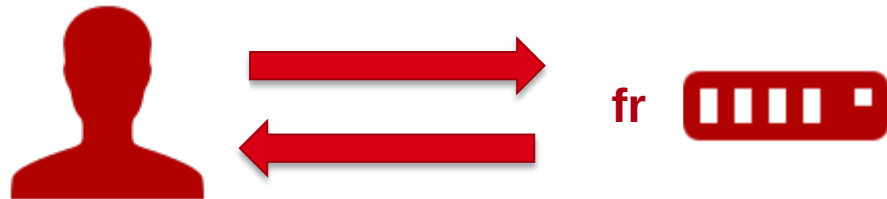
http://fr.dbpedia.org/data/Paris.rdf

fr.dbpedia.org/data/Paris.rdf

```
<rdf:Description rdf:about="http://fr.dbpedia.org/resource/Paris">
  <dbpedia-owl:wikiPageWikiLink rdf:resource="http://fr.dbpedia.org/resource/Paris"/>
</rdf:Description>
<rdf:Description rdf:about="http://fr.dbpedia.org/resource/Stéphane_Mallarmé">
  <dbpedia-owl:birthPlace rdf:resource="http://fr.dbpedia.org/resource/Paris"/>
  <dbpedia-owl:wikiPageWikiLink rdf:resource="http://fr.dbpedia.org/resource/Paris"/>
</rdf:Description>
<rdf:Description rdf:about="http://fr.dbpedia.org/resource/Système_international_d'unités">
  <dbpedia-owl:wikiPageWikiLink rdf:resource="http://fr.dbpedia.org/resource/Paris"/>
</rdf:Description>
<rdf:Description rdf:about="http://fr.dbpedia.org/resource/Sharon_Stone">
  <dbpedia-owl:wikiPageWikiLink rdf:resource="http://fr.dbpedia.org/resource/Paris"/>
</rdf:Description>
<rdf:Description rdf:about="http://fr.dbpedia.org/resource/Seine">
  <prop-fr:villes rdf:resource="http://fr.dbpedia.org/resource/Paris"/>
  <dbpedia-owl:city rdf:resource="http://fr.dbpedia.org/resource/Paris"/>
  <dbpedia-owl:wikiPageWikiLink rdf:resource="http://fr.dbpedia.org/resource/Paris"/>
</rdf:Description>
<rdf:Description rdf:about="http://fr.dbpedia.org/resource/Seconde_Guerre_mondiale">
  <dbpedia-owl:wikiPageWikiLink rdf:resource="http://fr.dbpedia.org/resource/Paris"/>
</rdf:Description>
<rdf:Description rdf:about="http://fr.dbpedia.org/resource/Stanley_Kubrick">
  <dbpedia-owl:wikiPageWikiLink rdf:resource="http://fr.dbpedia.org/resource/Paris"/>
</rdf:Description>
<rdf:Description rdf:about="http://fr.dbpedia.org/resource/Sigmund_Freud">
```

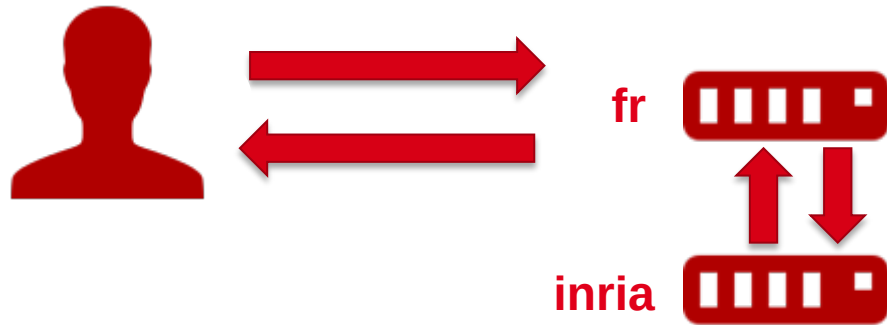
Importance of the Domain Name

`http://ns.inria.fr/fabien.gandon#me`



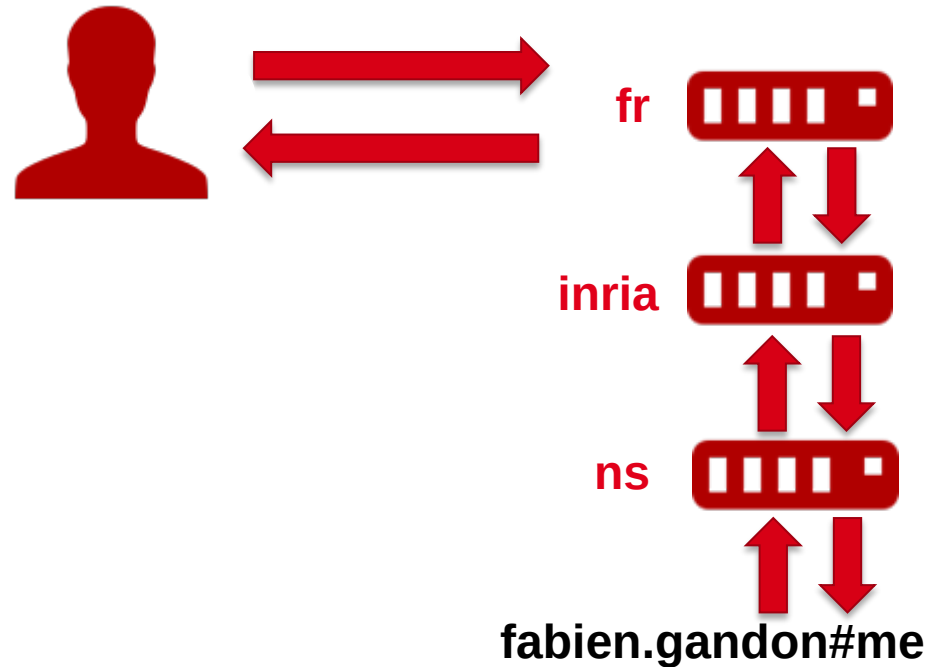
Importance of the Domain Name

`http://ns.inria.fr/fabien.gandon#me`



Importance of the Domain Name

`http://ns.inria.fr/fabien.gandon#me`



ratatouille.fr

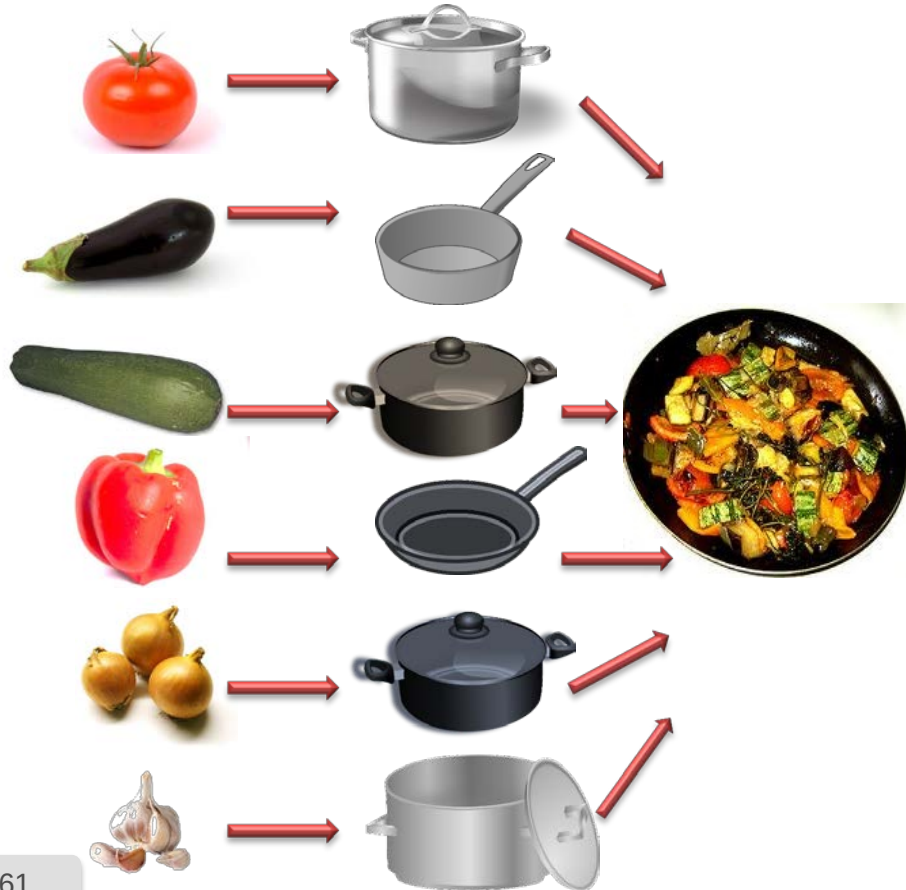
or the recipe for linked data



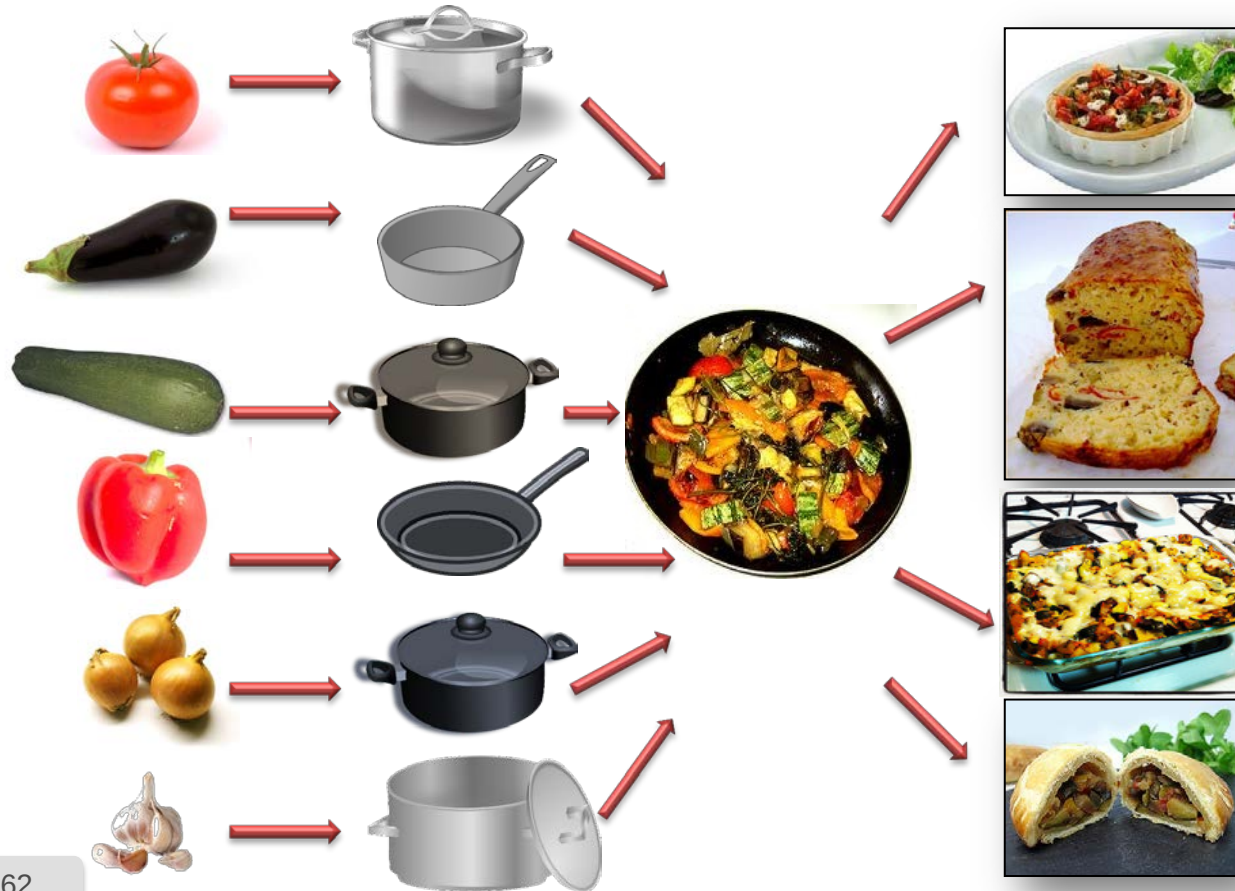
or the recipe for linked data

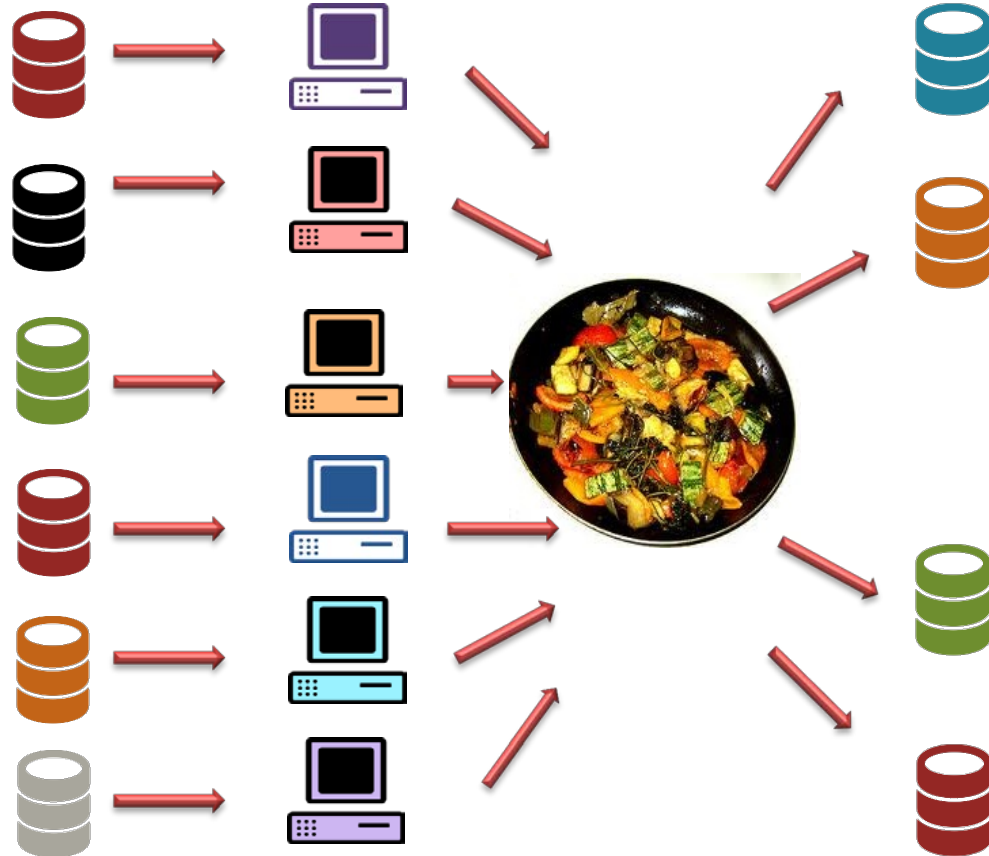


or the recipe for linked data

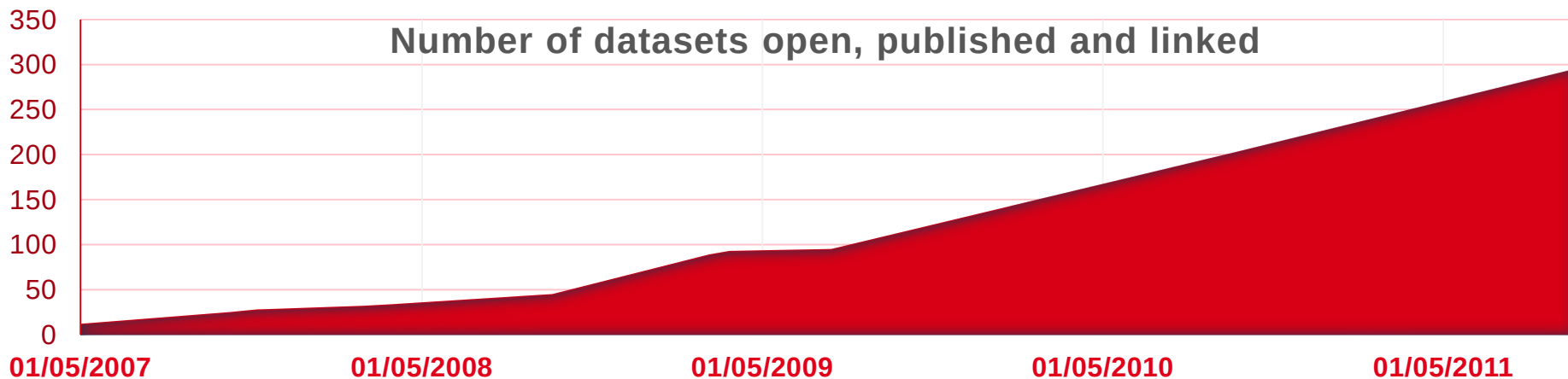
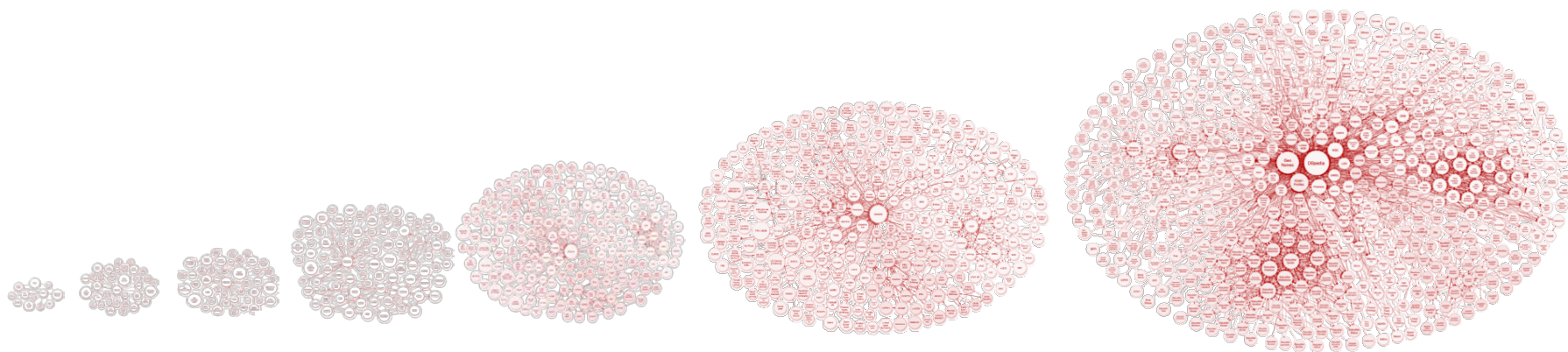


or the recipe for linked data





Linked Open Data Explosion on the Web



Picture credits

- 5 Star Linked Open Data mug, W3C, http://www.cafepress.com/w3c_shop.597992118
- Screenshots of Dbpedia – Paris – fr.dbpedia.org/page/Paris and fr.dbpedia/data/Paris.rdf
- vegetable-371919_640 by elementus – CC0 Public Domain <http://pixabay.com/fr/l%C3%A9gumes-red-tomate-371919>
- aubergine-1809_640 by PublicDomainPictures –CC0 Public Domain <http://pixabay.com/fr/aubergine-bio-r%C3%A9gime-alimentaire-1809/>
- zucchini-176880_640 by Maddox74 –CC0 Public Domain <http://pixabay.com/fr/courgette-alimentation-sant%C3%A9-manger-176880/>
- bell-pepper-390005_640 by PDPics – CC0 Public Domain <http://pixabay.com/fr/poivron-piment-doux-capsicum-red-390005/>
- onions-74355_640 by Lebensmittelfotos – CC0 Public Domain <http://pixabay.com/fr/oignons-l%C3%A9gumes-ingr%C3%A9dient-74355/>
- garlic-220495_640 by Kaz – CC0 Public Domain <http://pixabay.com/fr/l-ail-ampoule-clou-de-girofle-220495/>
- frying-pan-147684_640 by OpenClips – CC0 Public Domain <http://pixabay.com/fr/po%C3%A4le-%C3%A0-fr%C3%A4re-po%C3%A4le-cuisson-cuisine-147684/>
- pan-37803_640 by Nemo –CC0 Public Domain <http://pixabay.com/fr/pan-friture-cuisine-cuisson-37803/>
- cooking-pot-146459_640 by OpenClips –CC0 Public Domain <http://pixabay.com/fr/la-marmite-casserole-pot-cuisson-146459/>

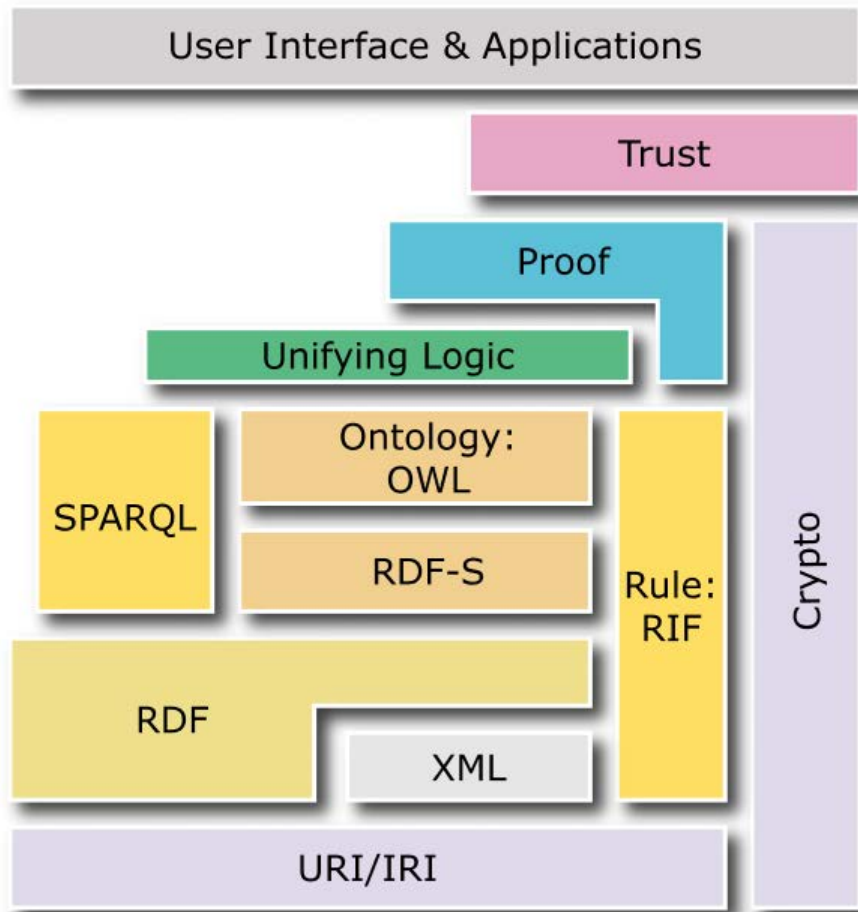
Picture credits

- cooking-pot-159470_640 by OpenClips – CC0 Public Domain <http://pixabay.com/fr/la-marmite-cook-ware-cuisini%C3%A8re-159470/>
- pan-156154_640 by OpenClips – CC0 Public Domain <http://pixabay.com/fr/pan-pot-cuisson-alimentaire-156154/>
- Ratatouille by marcurg – CC BY 2.0 <https://flic.kr/p/3egC98>
- Roasted Ratatouille Lasagna - Brent Pliskow – CC BY NC ND <https://flic.kr/p/baUxVP>
- Tartelettes à la ratatouille Par Philadelphia <http://www.750g.com/tartelettes-a-la-ratatouille-r53524.htm>
- Cake à la ratatouille, Du sel dans ma cuisine, 27/05/2012 <http://selengrain.canalblog.com/archives/2012/05/27/24295313.html>
- Chausson farci à la ratatouille, La Pâtisserie Moderne, 02/04/2014 <http://chef-chater.blogspot.fr/2014/04/phases-techniques-de-la-recette-1-pour.html>
- Linking Open Data cloud diagrams by Max Schmachtenberg, Christian Bizer, Anja Jentzsch and Richard Cyganiak. <http://lod-cloud.net/>
- Icons (iconfinder.com)
 - Global, globe, internet icon, iOS 7 icons by Visual Pharm – CC BY ND 3.0
 - Person icon, Ionicons by Ionicons - MIT License
 - Location icon, Ionicons by Ionicons – MIT Licence
 - Cog, configuration, gear, gears, machine part icon by Popcic – Free for commercial use
 - Server icon, Octicons by Github - MIT License

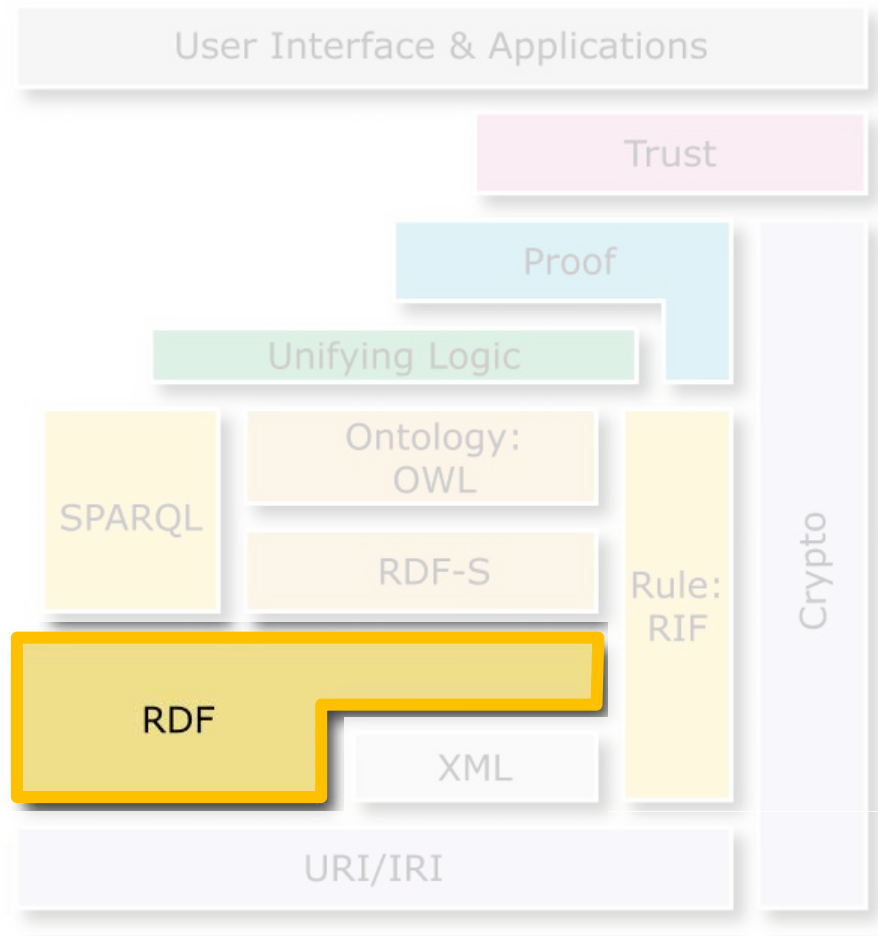
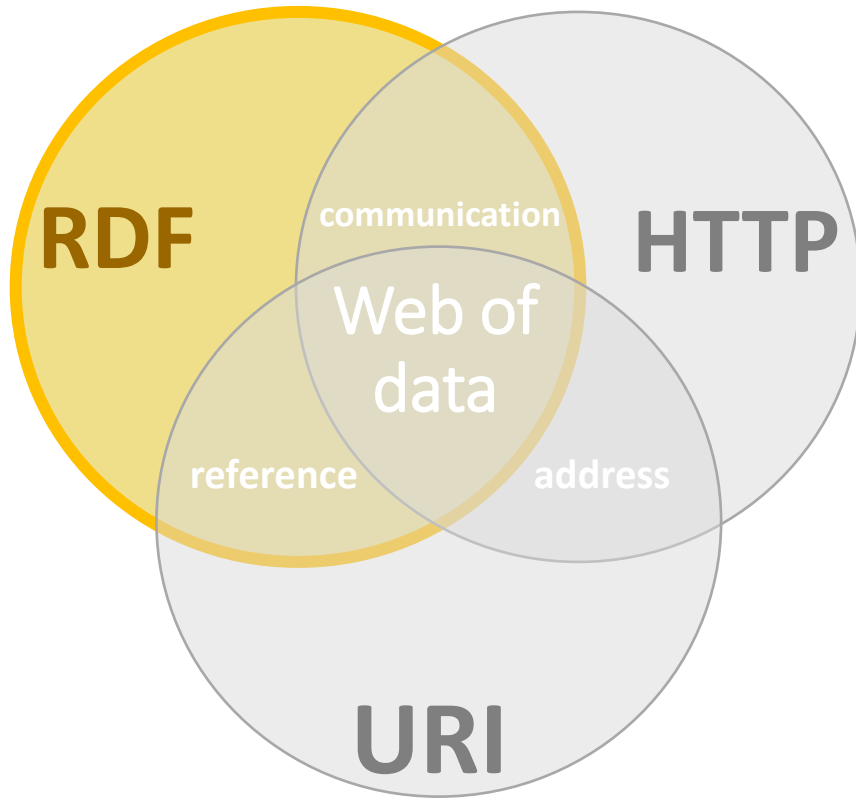
WEEK 1: Principles of a Web of Linked Data

1. Web architecture
2. Separating presentation and content
3. From pages to resources
4. Linked data principles
- 5. Stack of standards and languages**

W3C standards



W3C standards



RDF data model

<http://ns.inria.fr/fabien.gandon#me>

<http://inria.fr/schema#author>

<http://inria.fr/rr/doc.html>

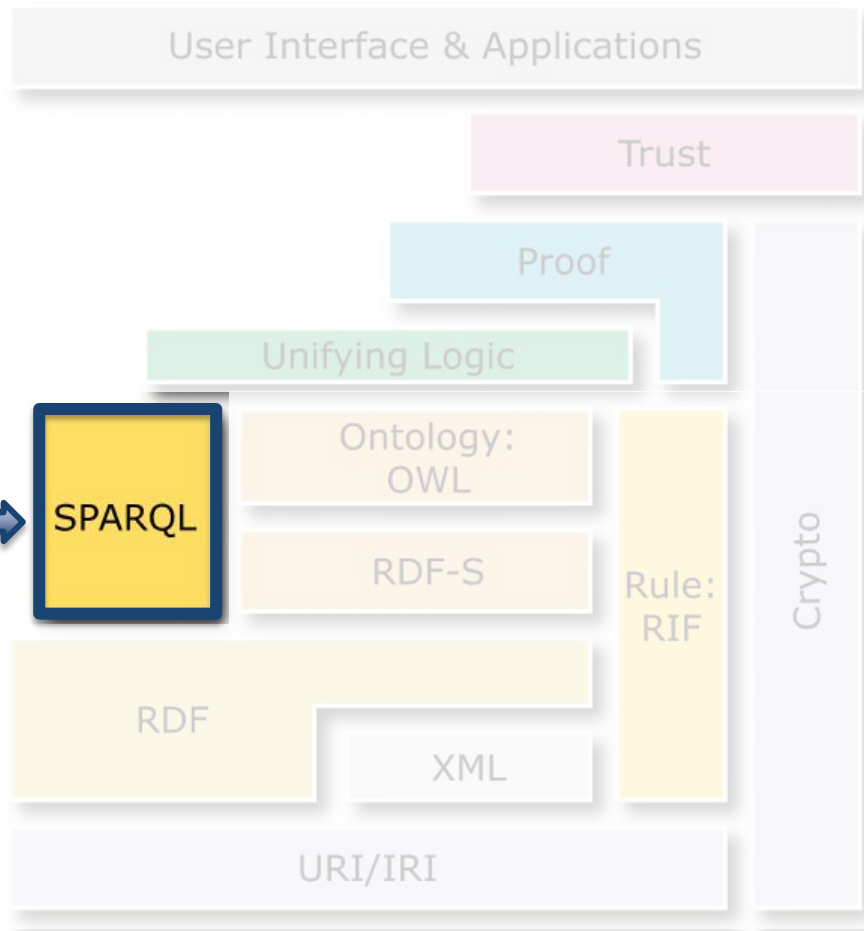
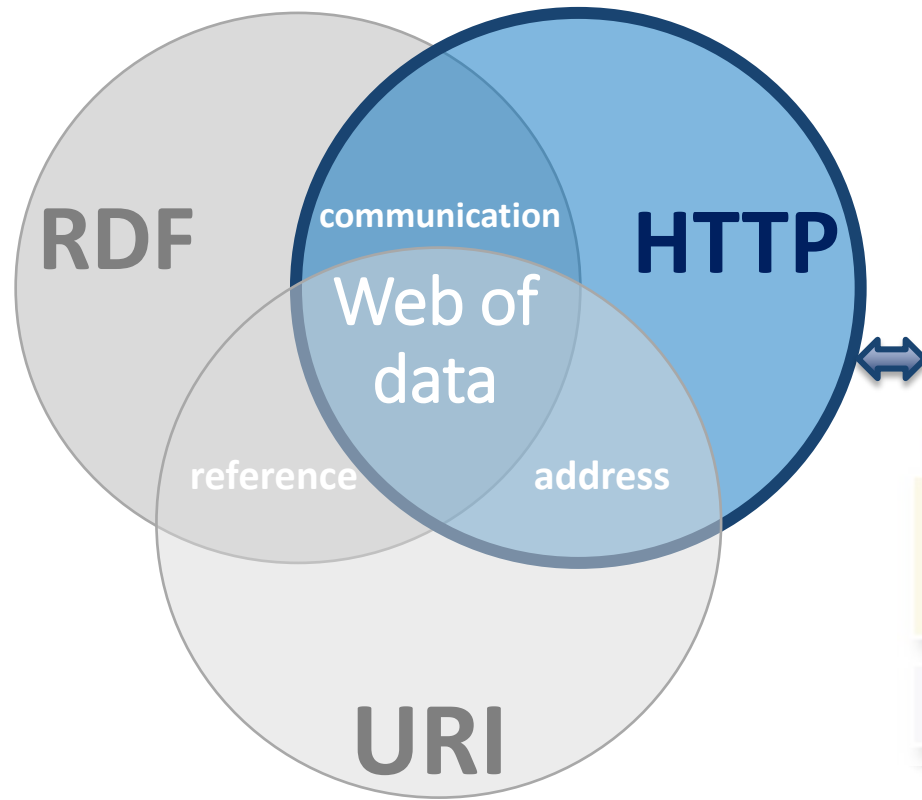
<http://inria.fr/rr/doc.html>

<http://inria.fr/schema#topic>

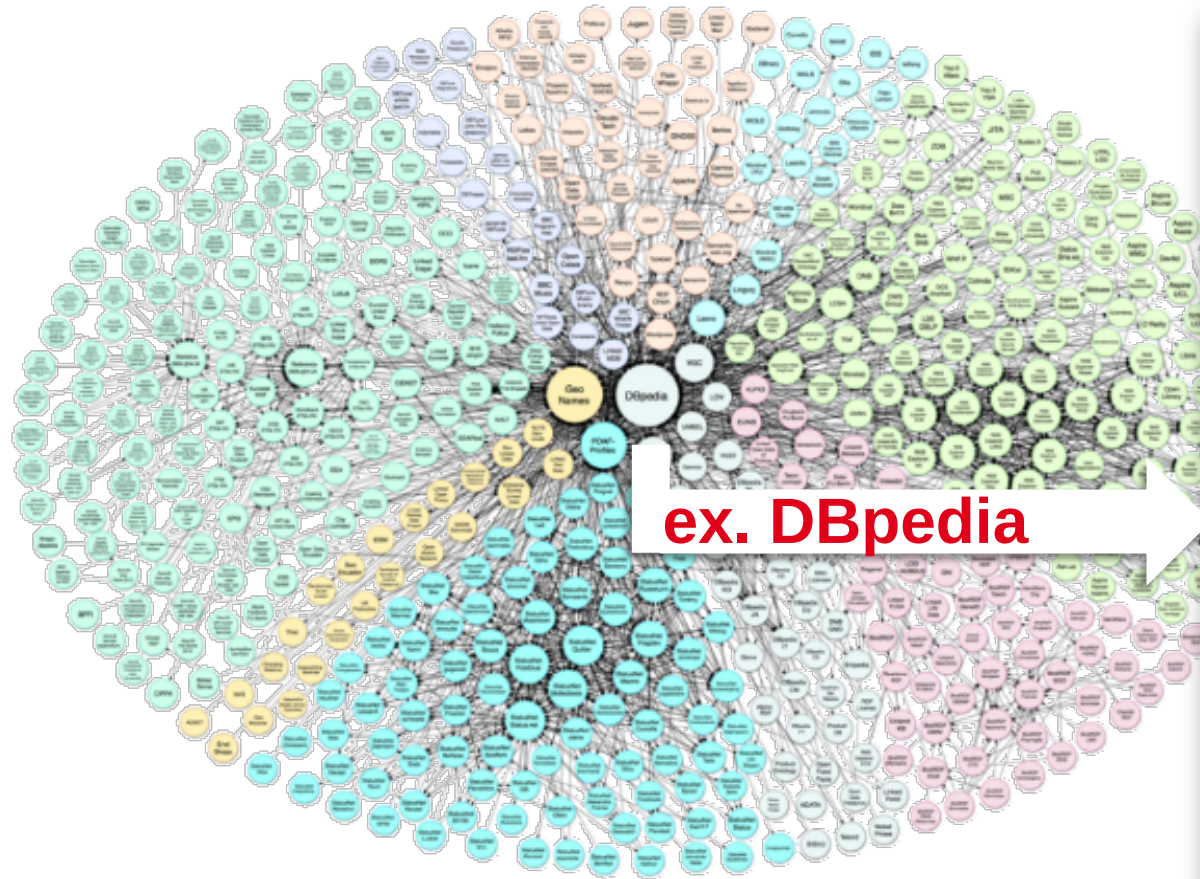
<http://inria.fr/schema#keyword>

"Music"

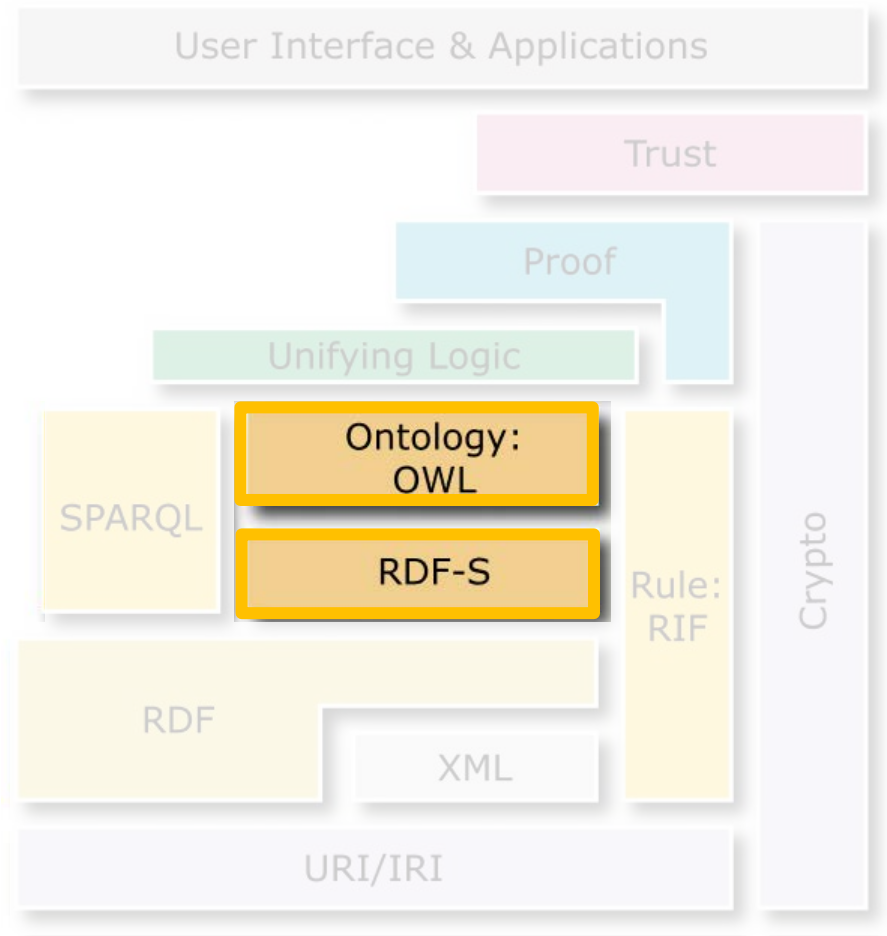
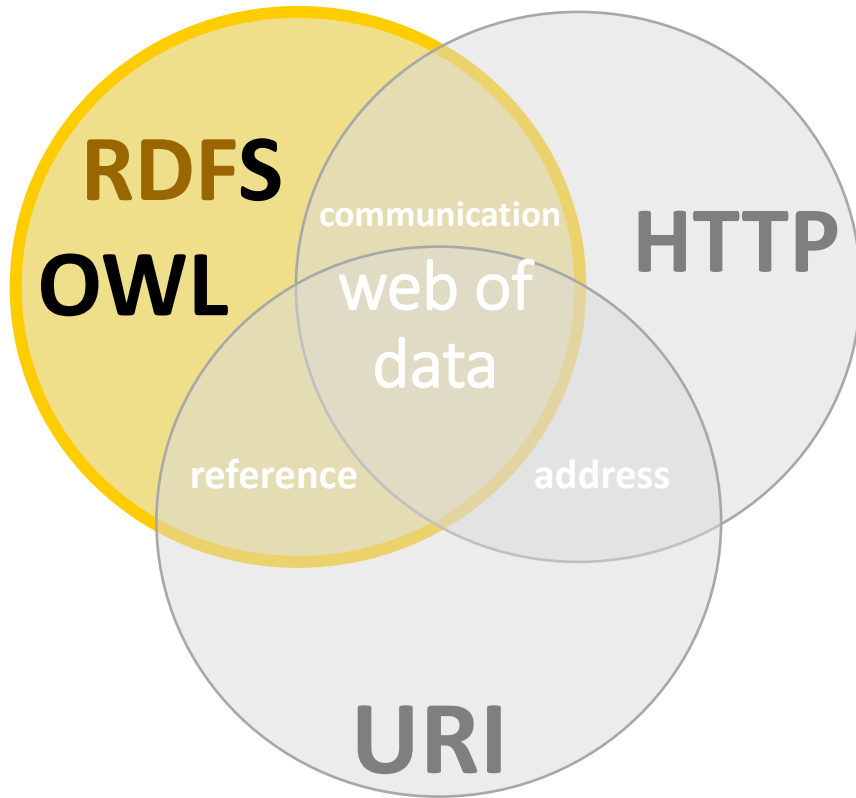
W3C standards



Get Data, Not Documents



W3C standards

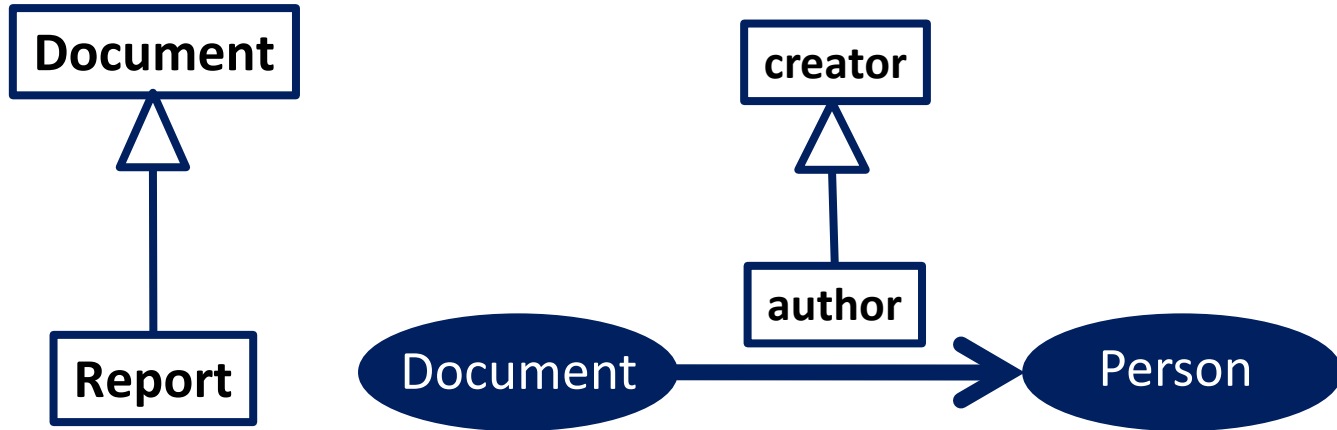


Ontologies, Schemas, Vocabularies... Outside the Scope

- Define the primitives / terms used in the descriptions and provide their semantics using formal languages
- Published at a namespace URI to provide the definitions of the names

RDFS

to declare classes of resources,
properties, and organize their hierarchy

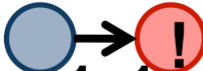


OWL in one... slide

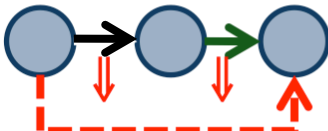
 algebraic properties



 disjoint properties

 qualified cardinality

 individual prop. neg

 chained prop.



union



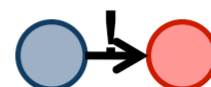
disjunction



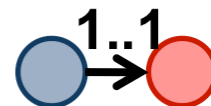
intersection



complement



restriction



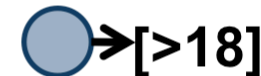
cardinality



equivalence



enumeration



value restriction

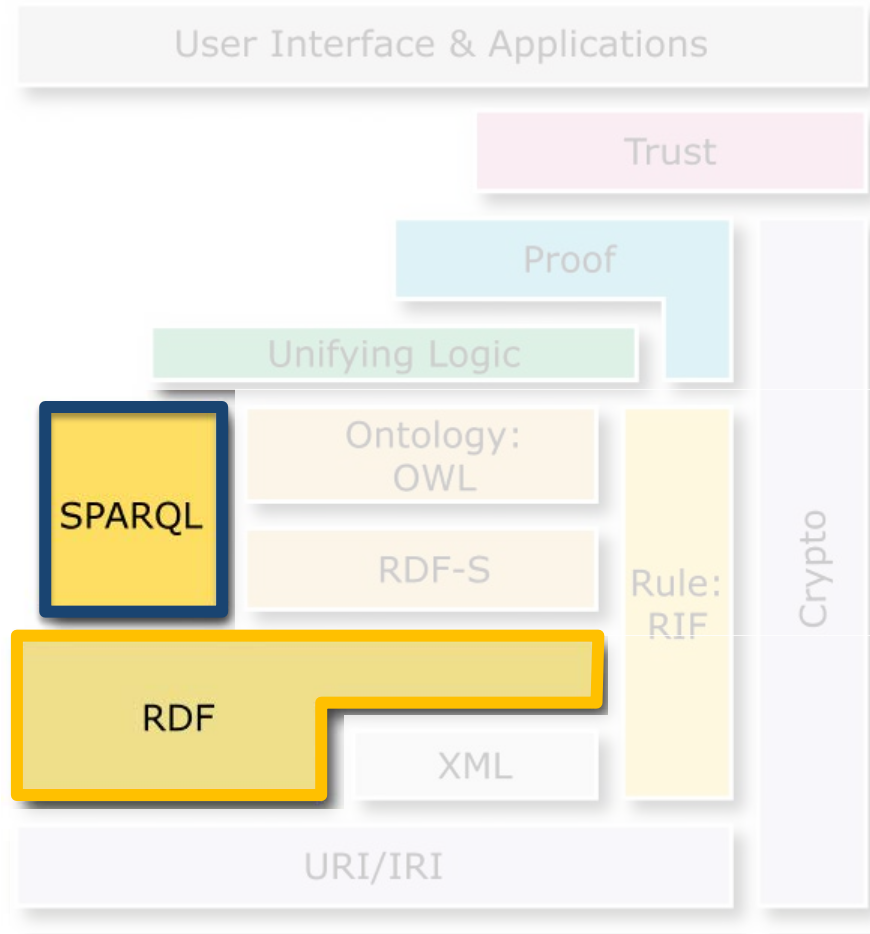


disjoint union



keys

In the Next Weeks...



Picture credits

- Semantic Web stack of standards, W3C®
- Linking Open Data cloud diagram by Max Schmachtenberg, Christian Bizer, Anja Jentzsch and Richard Cyganiak. <http://lod-cloud.net/>
- Icons
 - Database icon, Octicons by Github - MIT License (iconfinder.com)
 - Key icon, by Github – MIT Licence (iconfinder.com)
 - Server icon, Icocentre Free Icons by IcoCentre – Free for commercial use
 - Desktop computer symbol, Wpclipart.com - Public domain